

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000815	Version:	4.0, CURRENT
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DG-57 / DG-517 Service Manual

CAMERAS MODULE VERIFICATION

Procedure description
This procedure describes how to check the Cameras module: Reagents, Samples and Reader cameras.

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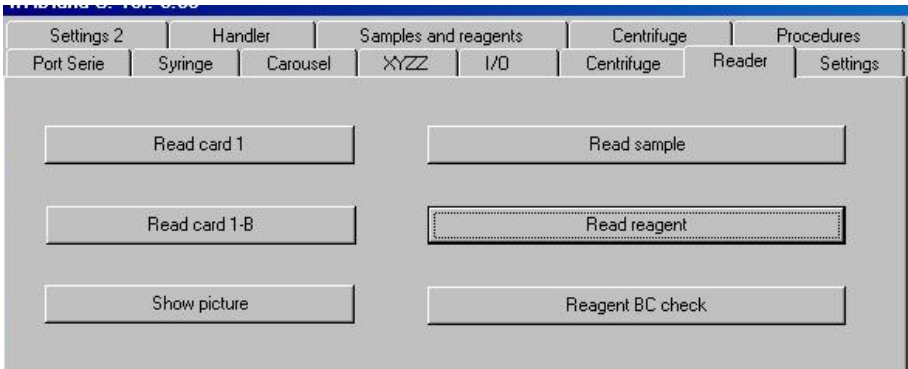
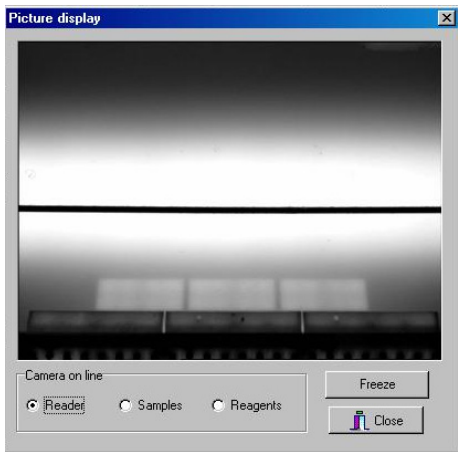
EQUIPMENT

1. An instrument connected with a PC with the software installed

SET UP

1. Turn on the instrument.
2. Execute the ST software (Technical Service software).
3. Enter the Technical Service password (d+g_diana).

PROCEDURE 1 - Reagents camera

Process	Expected Results
1. Access 'Reader' tab in ST software.  Picture 1	The software looks like Picture 1.
2. Press the 'Show picture' button. (see Picture 1)  Picture 2	The Picture 2 window appears.

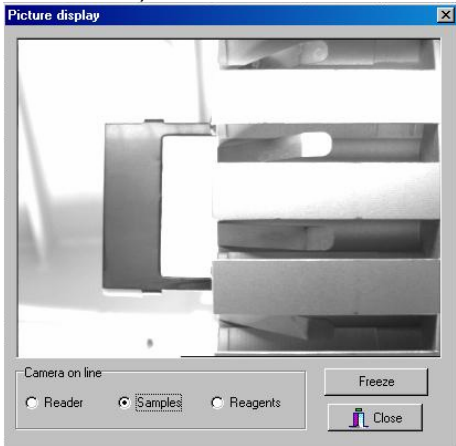
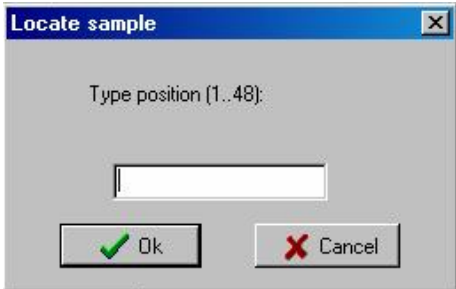
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Process	Expected Results
3. Select 'Reagents'. (See Picture 2). Picture 3 4. Check that image is present. (No black image) 5. If no image appears (black image) check the video cables, check the 12 volts power and check the camera itself. 6. Press the 'Close' button 7. Place a Vial with a barcode in position 1 of the reagent carousel. 8. Press the 'Reagent BC check' button (see picture 1). Picture 4 9. Type 1 and press the 'Ok' button. Picture 5 10. Type the bar code of the vial and press the 'Start' button. 11. Wait until 1000 readings are made. 12. Check the number of 'fails' is 0 and the number of 'Not detected' is less than 5.	The Picture 3 window appears. --- --- Picture 3 window closes. --- Picture 4 window appears. --- Position 1 of the reagents carousel places in reading position and Picture 5 window appears. --- The instrument starts reading the barcode. --- ---

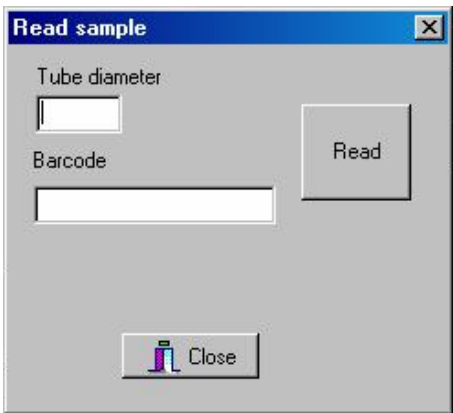
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Process	Expected Results
13. Otherwise check first the spinning system (see document REGD-0000797) and then the camera iris and mainly the camera focus (See document REGD-0000811).	---

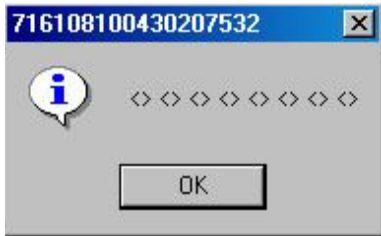
PROCEDURE 2 - Samples camera

Process	Expected Results
1. Access 'Reader' tab in ST software. 2. Press the 'Show picture' button. (see Picture 1) 3. Select 'Samples'. (See Picture 2).	The software looks like Picture 1. The Picture 2 window appears.
 Picture 6	The Picture 6 window appears.
4. Check that image is present. (No black image) 5. If no image appears (black image) check the video cables, check the 12 volts power and check the camera itself. 6. Press the 'Close' button	--- --- Picture 6 window closes.
7. Place a tube with a user's barcode in position 1 of the samples carousel. 8. Press the 'Read sample' button.	--- The Picture 7 window appears.
 Picture 7	

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Process	Expected Results
9. Type 1 and press the 'Ok' button.  Picture 8	Picture 8 window appears.
10. Press the 'Read' button.	---
11. Check that the 'Tube diameter' and the 'Barcode' are correct.	---
12. If 'Tube diameter' is not correct, check the samples camera alignment. (See document REGD-0000810).	---
13. If 'Barcode' can't be read, check the samples camera focus and iris. (See document REGD-0000810).	---

PROCEDURE 3 - Reader camera

Process	Expected Results
1. Access 'Reader' tab in ST software.	The software looks like Picture 1.
2. Press the 'Show picture' button. (see Picture 1)	The Picture 2 window appears.
3. Select 'Reader'. (See Picture 2)	---
4. Check that image is present. (No black image)	---
5. If no image appears (black image) check the video cables, check the 12 volts power and check the camera itself.	---
6. Press the 'Close' button	Picture 2 window closes.
7. Place a Neutral Card in the first position of the incubator.	---
8. Press the 'Read card 1' button.	Instrument picks up the card and reads it. Then Picture 9 window appears.
 Picture 9	
9. Check the barcode is correct and all results are '<>'.	---
10. If can't read the barcode, check the reader camera alignment and focus. (see document REGD-0000812).	---
11. If the well results are not '<>', check the neutral card, the reader camera iris and the reference image (see documents REGD-0000812 and REGD-0000814).	---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	Document REGD-0000815_v2 replaces document 57042V0M000	Uploaded document to GxPharma.
3.0	2011/04/28	Updated all the pictures.	From software version 3.x, the installation program of Diana System does not change the Colour Register of the Windows Register.
3.0	2011/04/28	Updated the Document Change History: Date format changed to YYYY/MM/DD Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	See document header.	Document title.	Commercial reference substituted by project reference.
4.0	See document header.	Some pictures added to easy understanding.	Procedure updated.
4.0	See document header.	Pictures updated.	Pictures updated with last software version pictures.
4.0	See document header.	Document header.	Increase information to easy document search.